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Studierichting: Informatica
Oefeningenreeks 6B nr. 17.3

$$
\begin{aligned}
s_n &= \sum_{k=1}^{n} k(k+1)(k+2) \\
&= \sum_{k=1}^{n} k\left(k^2+3k+2\right) \\
&= \sum_{k=1}^{n} k^3+3k^2+2k \\
&= \sum_{k=1}^{n} k^3 + 3\sum_{k=1}^{n} k^2 + 2\sum_{k=1}^{n} k \\
&= \frac{n^2\left(n+1\right)^2}{4} + 3\cdot\frac{n\left(n+1\right)\left(2n+1\right)}{6} + 2\cdot\frac{n\left(n+1\right)}{2} \\
&= \frac{n^2\left(n+1\right)^2}{4} + \frac{n\left(n+1\right)\left(2n+1\right)}{2} + n\left(n+1\right) \\
&= \frac{n\left(n+1\right)\left(n\left(n+1\right)+2\left(2n+1\right)+4\right)}{4} \\
&= \frac{n\left(n+1\right)\left(n^2+n+4n+2+4\right)}{4} \\
&= \frac{n\left(n+1\right)\left(n^2+5n+6\right)}{4} \\
&= \frac{n\left(n+1\right)\left(n+2\right)\left(n+3\right)}{4}
\end{aligned}
$$

Dus de algemene term van s_n is altijd geheel omdat:

$$n\left(n+1\right)\left(n+2\right)\left(n+3\right) \mod 4 = 0$$

References